

Exome Sequencing in 10 Unrelated Pedigrees with Autosomal-Recessive Inherited Retinal Degeneration Identifies Both Known and Novel Causative Variants and Highlights Multi-Gene Attribution Challenges

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Abstract

Purpose. To identify causative mutations and characterise phenotype–genotype association in 10 unrelated families segregating autosomal-recessive inherited retinal degeneration (arIRD). **Methods.** Probands and available relatives from 10 unrelated pedigrees (F1–F10) underwent standardised ophthalmic evaluation. Whole-exome sequencing (WES) was performed in eight pedigrees (F1–F8). Two additional pedigrees of self-reported Ashkenazi Jewish ancestry (F9, F10) underwent targeted Sanger and amplicon-based sequencing of DHDDS, based on the prevalence of the c.124A>G founder allele in this population. Candidate variants were filtered by minor-allele frequency, predicted impact, and known disease association; segregation was assessed in available family members; and variants absent from public databases were screened in a cohort of 100 ethnicity-matched control individuals (200 chromosomes). **Results.** A molecularly diagnostic or strongly candidate variant configuration was identified in all 10 pedigrees. Implicated genes included CERKL, ABCA4, RPE65, ARL6, USH2A, PDE6B and DHDDS. Eight of 10 pedigrees (F1–F6, F9, F10) showed full segregation of the implicated variant(s) with disease. In two pedigrees (F7, F8), potentially disease-causing variants were observed in *more than one* gene, complicating attribution of the primary cause. Histopathology of a donor eye from an F10 individual homozygous for DHDDS c.124A>G (p.Lys42Glu) demonstrated severe degeneration of retinal layers with relative preservation of the retinal pigment epithelium. **Conclusions.** Exome sequencing in unrelated arIRD pedigrees identifies both previously reported and novel candidate variants and, in a minority of pedigrees, candidate variants in more than one gene — creating challenges for causal attribution, genetic counseling, and prioritisation of gene-specific therapies. [Abstract, page 1]

Keywords: inherited retinal degeneration; exome sequencing; autosomal recessive; DHDDS; CERKL; ABCA4; RPE65; ARL6; USH2A; PDE6B; phenotype–genotype; Ashkenazi Jewish.

1. Introduction

Inherited retinal degenerations (IRDs) comprise a clinically and genetically heterogeneous group of monogenic disorders with an estimated combined prevalence in the range of 1 in 2,000 to 1 in 3,000. Autosomal-recessive IRD (arIRD) accounts for a substantial proportion of cases in consanguineous and population-isolate cohorts. Next-generation sequencing — particularly whole-exome sequencing (WES) — has accelerated molecular diagnosis and the discovery of novel disease genes. In parallel, gene-specific therapies (for example, subretinal AAV-mediated *RPE65* augmentation) have transformed counseling and prioritisation around precise molecular diagnosis. [§1, page 2]

In this report we present the molecular and clinical findings from 10 unrelated pedigrees with apparent arIRD. Our objective is two-fold: (i) to identify causative mutations across the cohort, and (ii) to characterise phenotype–genotype association at the level of the individual pedigree. [§1, page 2]

2. Materials and Methods

2.1 Patient ascertainment and ophthalmic evaluation

Ten unrelated probands with a clinical diagnosis of arIRD were ascertained between 2012-03 and 2016-09 from the Northbridge Eye Institute and the Beth-Mercy Center for Inherited Retinal Diseases. Pedigrees were designated F1 through F10. Each proband and available relative underwent best-corrected visual acuity (BCVA) measurement, slit-lamp biomicroscopy, dilated fundus examination, full-field electroretinography (ffERG) according to ISCEV standards, spectral-domain optical coherence tomography (SD-OCT), and short-wavelength fundus autofluorescence (FAF). Longitudinal follow-up was available in 7 of 10 pedigrees for ≥ 36 months. Family history was recorded across at least three generations where possible. [Methods §2.1, page 2]

2.2 DNA isolation

Genomic DNA was isolated from 8 mL peripheral whole blood collected in EDTA tubes using a salting-out procedure (Puregene; Qiagen). DNA quality was confirmed by 260/280 ratio (1.78–1.92 across samples) and quantified by Qubit dsDNA HS assay. Average yield was 44.6 μ g per sample (range 28.1–69.4 μ g). [Methods §2.2, page 2]

2.3 Exome capture and sequencing (pedigrees F1–F8)

For pedigrees F1–F8, 1 μ g of genomic DNA per proband was sheared (Covaris E220) to a median fragment length of 230 bp. Exome enrichment was performed using the Agilent SureSelect XT Human All Exon V5+UTR (75 Mb target) kit. Paired-end 2 $\times$ 100 bp sequencing was performed on Illumina HiSeq 2500. Mean on-target coverage across the eight probands was 96.4 $\times$ (range 88.2 $\times$ – 112.1 $\times$). Reads were aligned to GRCh37/hg19 with BWA-MEM v0.7.12; duplicates were marked with Picard v2.1.1; variants were called with GATK v3.5 (HaplotypeCaller) and annotated with ANNOVAR (build 2016-02-01) and VEP v83. [Methods §2.3, page 3]

2.4 Targeted DHDDS analysis (Ashkenazi Jewish pedigrees F9, F10)

Pedigrees F9 and F10 self-identified as Ashkenazi Jewish. Given the well-documented founder effect of *DHDDS* c.124A>G (p.Lys42Glu) in this population, all *DHDDS* coding exons and ± 20 bp flanking intronic sequence were amplified by 14 PCR amplicons and sequenced bidirectionally by Sanger sequencing (BigDye Terminator v3.1). Targeted amplicon-based deep sequencing was additionally performed on Illumina MiSeq with mean amplicon coverage 4,820 $\times$ across F9 and F10 probands. [Methods §2.4, page 3]

2.5 Variant filtering

WES variants were filtered by: (i) minor-allele frequency (MAF) < 0.005 in 1000 Genomes Phase 3, ExAC r0.3, and an in-house cohort of 1,184 exomes; (ii) presence in a manually curated panel of 281 IRD-associated genes; (iii) predicted moderate or high impact (missense, nonsense, frameshift, canonical splice $\pm 1/\pm 2$, in-frame indels); and (iv)

zygosity consistent with an autosomal-recessive model (homozygous or compound heterozygous). Synonymous and deep-intronic variants were excluded unless segregation or splice-prediction evidence flagged them. *[Methods §2.5, page 3]*

2.6 Segregation analysis

All candidate variants surviving §2.5 were validated by bidirectional Sanger sequencing in the proband and in every available first-degree relative. Segregation was scored as **confirmed** when the genotype distribution in the pedigree was consistent with the autosomal-recessive model across all sampled affected and unaffected individuals; **partial** when one or more expected relatives could not be sampled but the available genotypes were consistent; and **ambiguous** when candidate variants in distinct genes each segregated partially with disease, precluding a single causal call. *[Methods §2.6, page 3]*

2.7 Ethnicity-matched control analysis

All variants that were absent from 1000 Genomes, ExAC, and our in-house exome set ("novel" variants) were screened for presence in **100 ethnicity-matched control individuals (200 chromosomes)** by allele-specific PCR and Sanger confirmation. The control cohort was assembled to match the self-reported ethnicity of the corresponding proband (Ashkenazi Jewish for F9/F10; Northern European for F1, F2, F3, F8; South Asian for F4, F7; Maghrebi for F5; and Eastern European for F6). *[Methods §2.7, page 3]*

3. Results

3.1 Cohort summary

The cohort comprised 10 unrelated probands (6 female, 4 male) with a mean age at ascertainment of 27.4 years (range 4–58). Pedigrees F1–F8 underwent WES; F9 and F10 underwent targeted DHDDS analysis. A molecularly diagnostic or strongly candidate variant configuration was identified in **10/10** pedigrees. Segregation was **confirmed** in 8/10 pedigrees (F1, F2, F3, F4, F5, F6, F9, F10), **partial** in F7, and **ambiguous** in F8 due to candidate variants in two distinct genes. [§3.1, page 4]

Table 1. Per-pedigree summary of implicated gene(s), variant(s), zygosity, novelty, and segregation outcome. [Table 1, page 4]

Ped.	Gene(s)	Variant(s) (HGVS c. / p.)	Zygosity	Known / Novel	Segregation
F1	CERKL	c.769C>T (p.Arg257*)	homozygous	known	confirmed
F2	ABCA4	c.5882G>A (p.Gly1961Glu); c.4139C>T (p.Pro1380Leu)	compound het	both known	confirmed
F3	RPE65	c.271C>T (p.Arg91Trp)	homozygous	known	confirmed
F4	ARL6	c.281T>C (p.Ile94Thr)	homozygous	novel	confirmed
F5	USH2A	c.2299delG (p.Glu767Serfs*21)	homozygous	known	confirmed
F6	PDE6B	c.1923_1969del (p.Lys641Asnfs*8)	homozygous	known	confirmed
F7	CERKL + USH2A	CERKL c.847C>T (p.Arg283*); USH2A c.11864G>A (p.Trp3955*)	both homozygous	CERKL known; USH2A novel	partial
F8	ABCA4 + RPE65	ABCA4 c.6079C>T (p.Leu2027Phe) / c.5461-10T>C; RPE65 c.95-2A>T (het carrier)	ABCA4 comp.het; RPE65 het	both reported	ambiguous
F9	DHDDS	c.124A>G (p.Lys42Glu)	homozygous	known founder (AJ)	confirmed
F10	DHDDS	c.124A>G (p.Lys42Glu)	homozygous	known founder (AJ)	confirmed

Of the eight pedigrees that underwent WES (F1–F8), six (F1, F2, F3, F4, F5, F6) yielded a single causally implicated gene with full segregation. In F7 and F8, candidate variants of moderate or high impact were identified in **more than one** IRD-associated gene; we discuss the attribution challenge in §4.2. The two Ashkenazi Jewish pedigrees (F9, F10) were homozygous for the previously reported founder allele DHDDS c.124A>G (p.Lys42Glu). [§3.1, Table 1, page 4]

3.2 Sequencing performance metrics

Per-pedigree sequencing metrics are summarised in Table 2. **Mapped on-target reads** are reported for the WES pedigrees (F1–F8) under the SureSelect XT V5+UTR capture and for the targeted-amplicon DHDDS analysis in the Ashkenazi pedigrees (F9, F10). **Off-target reads** denote those mapping outside the capture region (WES) or outside the 14 DHDDS amplicons (F9, F10). These numbers are provided to support reproducibility and downstream auditing. [§3.2, page 5]

Table 2. Per-pedigree mapped on-target and off-target read counts. WES pedigrees (F1–F8) use the SureSelect XT V5+UTR 75 Mb capture; targeted pedigrees (F9, F10) use the 14-amplicon DHDDS panel. [Table 2, page 5]

Pedigree	Assay	Mapped on-target reads	Off-target reads	Mean on-target depth
F1	WES (V5+UTR)	87,412,336	412,336	92.4×
F2	WES (V5+UTR)	91,005,217	505,217	101.7×
F3	WES (V5+UTR)	84,777,902	377,902	88.2×
F4	WES (V5+UTR)	96,233,481	633,481	112.1×
F5	WES (V5+UTR)	88,119,654	419,654	94.0×
F6	WES (V5+UTR)	92,864,710	864,710	103.6×
F7	WES (V5+UTR)	90,551,028	551,028	97.8×
F8	WES (V5+UTR)	89,703,445	703,445	98.7×
F9	Targeted DHDDS (14 amp.)	1,204,886	24,886	4,612×
F10	Targeted DHDDS (14 amp.)	1,318,507	38,507	5,028×
—	Cohort total	723,191,166	4,531,166	—

Across the eight WES pedigrees the sum of mapped on-target reads was **720,667,773**. Across the two targeted-amplicon DHDDS pedigrees (F9, F10) the sum of mapped reads was **2,523,393**. The total off-target read count across all ten pedigrees was **4,531,166**. [§3.2, Table 2, page 5]

3.3 Control-cohort screening of novel variants

Two variants identified in our cohort were absent from 1000 Genomes Phase 3, ExAC r0.3, and our in-house 1,184-exome set, and were therefore considered **novel**: (i) ARL6 c.281T>C (p.Ile94Thr) in pedigree F4, and (ii) USH2A c.11864G>A (p.Trp3955*) in pedigree F7. Both novel variants were screened in **100 ethnicity-matched control individuals (200 chromosomes)** by allele-specific PCR with Sanger confirmation. **Neither novel variant was detected in any of the 100 ethnicity-matched controls**, supporting their rarity in the relevant background populations. [§3.3, page 6]

3.4 Phenotype over time, per pedigree

Table 3 summarises age at onset, presenting symptom, ffERG pattern, SD-OCT central macular thickness (CMT) at baseline and at the most-recent follow-up visit (where available), and the gene-associated disease pattern from the published literature. The observed phenotypes were **broadly consistent** with previously described gene-associated disease patterns in each pedigree. [§3.4, page 6]

Table 3. Per-pedigree quantitative phenotype summary. CMT = central macular thickness (µm); ffERG = full-field electroretinogram; rod-cone vs cone-rod refers to the dominant degeneration pattern. [Table 3, page 6]

Ped.	Gene	Age onset (y)	Presenting symptom	ffERG	CMT baseline / latest (µm)	Match to gene pattern
F1	CERKL	14	Bilateral central scotoma; pigmentary maculopathy	Severe cone-rod loss	228 / 162	Consistent: CERKL → cone-rod dystrophy
F2	ABCA4	11	Decreased central acuity; flecks	Cone > rod attenuation	256 / 184	Consistent: ABCA4 → Stargardt/CRD
F3	RPE65	4	Nyctalopia in early childhood	Non-detectable rod & cone	301 / 262	Consistent: RPE65 → LCA/EOSRD
F4	ARL6	9	Nyctalopia; mild polydactyly history	Severe rod-cone loss	274 / 199	Consistent: ARL6 → BBS3 / non-syndromic RP
F5	USH2A	17	Nyctalopia; mild hearing impairment	Rod-cone dystrophy	289 / 226	Consistent: USH2A → Usher II / arRP
F6	PDE6B	6	Early-onset nyctalopia	Non-detectable rod	262 / 188	Consistent: PDE6B → arRP
F7	CERKL+USH2A	13	Nyctalopia + central acuity loss	Mixed rod-cone & cone-rod	248 / 174	Mixed; see §4.2
F8	ABCA4+RPE65	8	Decreased acuity; nyctalopia	Generalised attenuation	271 / 205	Mixed; see §4.2
F9	DHDDS	10	Nyctalopia; midperipheral pigmentary changes	Severe rod-cone loss	283 / 211	Consistent: DHDDS → arRP59
F10	DHDDS	12	Nyctalopia; visual-field constriction	Severe rod-cone loss	279 / 196	Consistent: DHDDS → arRP59

3.5 Donor-eye histopathology — DHDDS p.Lys42Glu (pedigree F10)

A donor eye was obtained *post mortem* from an F10 individual homozygous for DHDDS c.124A>G (p.Lys42Glu) (age at death 63 y; clinical disease duration ~50 y). Fixed in 4% paraformaldehyde and processed for paraffin embedding, 5-µm sections were stained with haematoxylin–eosin (H&E) and processed for immunohistochemistry. Histopathology demonstrated **severe degeneration of the retinal layers** — including near-complete loss of the outer nuclear layer and outer segments, with attenuation of the inner nuclear and ganglion-cell layers in the periphery — and, notably, **relative preservation of the retinal pigment epithelium (RPE)** as a recognisable monolayer with retained pigment granules across the central and mid-peripheral retina. This pattern (severe neural-retinal degeneration with relative RPE preservation) is in keeping with the clinical picture of arRP59 attributable to DHDDS deficiency. [§3.5, page 7]

4. Discussion

4.1 Identification of known and novel variants

Across the eight WES pedigrees we identified causally implicated variants in six well-established arIRD genes (CERKL, ABCA4, RPE65, ARL6, USH2A, PDE6B) and one founder allele (DHDDS c.124A>G) in the two targeted Ashkenazi pedigrees. Two variants were **novel** — ARL6 c.281T>C (p.Ile94Thr) in F4 and USH2A c.11864G>A (p.Trp3955*) in F7 — and the remaining variants have been previously reported in association with the relevant disorders. We therefore explicitly **distinguish previously reported mutations from potentially pathogenic novel variants** in our pedigree-by-pedigree reporting (Table 1). [§4.1, page 8]

4.2 Multi-gene candidate variants in F7 and F8

In two pedigrees (F7, F8), our standard filtering pipeline returned potentially disease-causing variants in **more than one gene**, complicating attribution of the primary cause:

F7 carries a homozygous known truncating variant in *CERKL* (c.847C>T, p.Arg283*) *and* a homozygous novel truncating variant in *USH2A* (c.11864G>A, p.Trp3955*). Both genes are independently sufficient to cause arIRD. Segregation in the available siblings was consistent with either configuration; one second-degree relative was not available, precluding a definitive single-gene call. [§4.2, page 8]

F8 carries compound-heterozygous variants in *ABCA4* (c.6079C>T / c.5461-10T>C — both previously reported) *and* a heterozygous canonical-splice variant in *RPE65* (c.95-2A>T). Although a single heterozygous *RPE65* allele is not expected to cause recessive disease on its own, the co-occurrence of two candidate configurations in a single proband — combined with a mixed ERG pattern — precluded a confident *single-gene* causal call from genetic data alone. [§4.2, page 8]

These findings illustrate that even with stringent filtering and segregation analysis, exome sequencing may identify candidate variants in more than one gene per family, creating challenges for causal attribution, genetic counseling, and the selection of gene-specific therapies. [§4.2, page 8]

4.3 Phenotype–genotype correspondence

Where a single gene was implicated (F1–F6, F9, F10), the observed phenotype over time — age at onset, presenting symptom, fERG pattern, and SD-OCT progression — was broadly consistent with previously described gene-associated disease patterns (Table 3). For example, F3 (RPE65 p.Arg91Trp) presented with infantile nyctalopia and non-detectable fERG, consistent with the LCA/early-onset severe retinal dystrophy spectrum associated with *RPE65* loss-of-function. F2 (ABCA4 p.Gly1961Glu / p.Pro1380Leu) showed central-acuity decline and flecks in adolescence, consistent with Stargardt/cone-rod dystrophy. F5 (USH2A p.Glu767Serfs*21) showed rod-cone degeneration with mild hearing impairment, consistent with Usher syndrome type II. [§4.3, page 8]

4.4 Limitations

We did not perform functional validation of the two novel variants (ARL6 p.Ile94Thr; USH2A p.Trp3955*) within the scope of this study. The control cohort of 100 individuals (200 chromosomes) is small for absolute population-frequency claims; we report only that the novel variants were not observed in this cohort. Multi-generational parental DNA was not available in three of ten pedigrees, limiting the strength of segregation evidence in those families. [§4.4, page 9]

4.5 Conclusion

Exome sequencing can identify both known and novel disease-causing variants in unrelated arIRD pedigrees, and in a minority of pedigrees identifies candidate variants in more than one gene — creating challenges for causal attribution, genetic counseling, and selection of gene-specific therapies. Pedigree-level reporting that preserves the known-versus-novel distinction and explicitly flags multi-gene attribution ambiguity is

essential for downstream clinical interpretation. [§4.5, page 9]

Appendix A. Per-pedigree clinical and genetic narratives

A.F1

F1 is a non-consanguineous Northern European pedigree with one affected proband (F1-II:2, female, age 24 at ascertainment) and one unaffected sibling. WES identified a homozygous nonsense variant in *CERKL* (c.769C>T, p.Arg257*), a previously reported pathogenic allele. Sanger segregation confirmed heterozygosity in both parents (F1-I:1, F1-I:2) and homozygous wild-type genotype in the unaffected sibling (F1-II:1). Visual acuity declined from 20/40 at age 22 to 20/120 at age 25. [§A.F1, page 10]

A.F2

F2 is a non-consanguineous Northern European pedigree with two affected siblings (F2-II:1, F2-II:3) and one unaffected sibling (F2-II:2). WES identified compound-heterozygous *ABCA4* variants c.5882G>A (p.Gly1961Glu) and c.4139C>T (p.Pro1380Leu) — both previously reported in Stargardt/CRD spectrum. Parental Sanger confirmed each parent contributed one allele in trans. [§A.F2, page 10]

A.F3

F3 is a consanguineous Northern European pedigree (first-cousin parents). The proband (F3-IV:1, male, age 4) presented with infantile nyctalopia. WES identified a homozygous *RPE65* missense variant c.271C>T (p.Arg91Trp), a known pathogenic allele. Both parents were heterozygous. ffERG was non-detectable for rod and cone responses. [§A.F3, page 10]

A.F4

F4 is a consanguineous South Asian pedigree. The proband (F4-IV:2, female, age 9) presented with nyctalopia and a maternal-side history of mild postaxial polydactyly (without obesity or renal involvement). WES identified a homozygous **novel** missense variant in *ARL6*: c.281T>C (p.Ile94Thr). The variant was absent from 1000 Genomes, ExAC, our in-house exome set, and from 100 ethnicity-matched South Asian control individuals (200 chromosomes). Heterozygosity was confirmed in both parents by Sanger. [§A.F4, page 11]

A.F5

F5 is a consanguineous Maghrebi pedigree. The proband (F5-III:1, male, age 17) showed rod-cone dystrophy with mild bilateral high-frequency hearing impairment on audiometry. WES identified the homozygous frameshift *USH2A* variant c.2299delG (p.Glu767Serfs*21), a recurrent Usher type II allele. [§A.F5, page 11]

A.F6

F6 is a consanguineous Eastern European pedigree. The proband (F6-III:2, female, age 6) showed early-onset nyctalopia and non-detectable rod ERG. WES identified a homozygous 47-bp deletion in *PDE6B*: c.1923_1969del (p.Lys641Asnfs*8), previously reported in arRP. [§A.F6, page 11]

A.F7

F7 is a consanguineous South Asian pedigree. The proband (F7-IV:1, male, age 13) showed mixed nyctalopia and central-acuity decline. WES returned **two** homozygous candidate truncating variants in distinct genes: *CERKL* c.847C>T (p.Arg283*, known) and *USH2A* c.11864G>A (p.Trp3955*, novel — absent from 100 ethnicity-matched South Asian controls). One critical second-degree relative (F7-III:4) was not available for sampling, so segregation was scored as **partial**; attribution between *CERKL* and *USH2A* could not be made on genetic data alone. [§A.F7, page 11]

A.F8

F8 is a non-consanguineous Northern European pedigree. The proband (F8-II:1, female, age 8) showed mixed nyctalopia and acuity loss. WES returned compound-heterozygous *ABCA4* variants c.6079C>T (p.Leu2027Phe) and c.5461-10T>C (both reported) **and** a heterozygous canonical-splice *RPE65* variant c.95-2A>T. A single heterozygous *RPE65* allele is not expected to cause recessive disease on its own, but combined with a mixed ERG pattern this configuration prevented a confident single-gene call. Segregation was scored as **ambiguous**. [SA.F8, page 12]

A.F9

F9 is a non-consanguineous Ashkenazi Jewish pedigree. The proband (F9-II:2, female, age 10) presented with nyctalopia and pigmentary midperipheral changes. Targeted DHDDS analysis identified the homozygous founder allele c.124A>G (p.Lys42Glu). Heterozygosity was confirmed in both parents. [SA.F9, page 12]

A.F10

F10 is a non-consanguineous Ashkenazi Jewish pedigree. The proband (F10-II:1, male, age 12) presented with nyctalopia and visual-field constriction. Targeted DHDDS analysis identified the homozygous founder allele c.124A>G (p.Lys42Glu). A donor eye from a deceased older relative (F10-I:2, homozygous for the same allele) was analysed histopathologically (see §3.5). [SA.F10, page 12]

Appendix B. Numeric audit of sequencing read counts

This appendix reprints the per-pedigree read counts from Table 2 in raw integer form, to facilitate downstream auditing. These same values are referenced verbatim in the Results §3.2 cohort totals. [§Appendix B, page 13]

Pedigree	Mapped on-target reads (integer)	Off-target reads (integer)
F1	87412336	412336
F2	91005217	505217
F3	84777902	377902
F4	96233481	633481
F5	88119654	419654
F6	92864710	864710
F7	90551028	551028
F8	89703445	703445
F9	1204886	24886
F10	1318507	38507
WES sum (F1..F8)	720667773	—
Targeted sum (F9..F10)	2523393	—
Off-target sum (F1..F10)	—	4531166

Control-cohort denominator: 2×100 ethnicity-matched controls = **200** chromosomes. Confirmed-segregation pedigrees per §3.1 / Table 1: **8** (F1, F2, F3, F4, F5, F6, F9, F10). [§Appendix B, page 13]

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